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Summary

Experienced Python Developer and Software Engineer with 16+ years of expertise in automation, embedded systems, and scalable software design. Skilled at modernising legacy systems, streamlining workflows, and building efficient, maintainable codebases. Equally effective as an independent problem-solver or collaborative team member, with a proactive, results-focused approach that consistently delivers measurable improvements in performance and reliability.

Key Skills

- **Python Development:** 16+ years building scalable, maintainable applications with advanced use of Python,
- **Automation & QA:** Expertise in test automation (PyTest, Robot Framework, Behave, Playwright, Selenium) and CI/CD integration for QA, networking, and DevOps workflows.
- **DevOps & Cloud:** CI/CD pipelines with GitHub Actions, GitLab CI, Jenkins; Infrastructure as Code (Ansible); containerization with Docker; cloud-native development.
- **Data & Systems:** ETL and data processing with Pandas/NumPy; NoSQL databases (CouchDB, Elasticsearch, Redis); microservice architecture design.
- **Networking & Linux:** Strong grounding in networking protocols/tools (Wireshark, tcpdump, IXIA, Spirent); Linux system administration and Bash scripting.
- **Collaboration & Documentation:** Experienced in open-source contributions, technical documentation, and mentoring.
- **IoT & Hardware:** Hands-on work with single-board computers and IoT device prototyping.

Employment History

Senior Scientific Software Engineer, MetOffice (Dec. 2024 – Jul. 2025)

- Designed and implemented automated workflows using Cylc to replace legacy data feeds, streamlining the extraction and processing of meteorological data from MOGREPS-UK, UKV, and Global weather models.
- Developed data pipelines in Microsoft Azure, leveraging cloud resources for rapid prototyping and scalability, before deploying the final production workflows to the MetOffice supercomputer environment for operational use.
- Worked extensively with NetCDF formats and used Python libraries like xarray, numpy, pandas and scipy for efficient handling and analysis of high-resolution, multidimensional datasets.
- Utilised Dask to parallelise and optimise performance for large-scale data processing tasks.
- Managed environments with Conda/Mamba and integrated JFrog Artifactory for secure, versioned package management; and implemented CI/CD pipelines using GitHub Actions.
- Followed Python best practices, including modular architecture, unit testing, version control (Git), documentation, and PEP 8 compliance to deliver robust and maintainable code.
- Liaised with stakeholders to gather requirements, ensuring new data services were aligned with operational needs and delivered measurable improvements in reliability and performance.
- Promoted high standards of scientific software development within the team, including mentoring colleagues and contributing to long-term technical strategy.

Python Developer, Freelance (Sep. 2023 – Nov. 2024)

- Provided direct independent software development services to several clients, with recent focus on start-up IoT projects, medical diagnostic devices and robotic SDK.

AMD, Senior Python Developer (Jan. 2020 – Jul. 2023)

- Led a team to design and implement a fully automated CI testing framework using FastAPI on AWS (EC2, S3, SQS, SNS) and MongoDB. Leveraged Python Lambda functions and Boto3 for orchestration.
- Automated the extensive test network infrastructure operations by utilising Python libraries like nornir, netmiko, paramiko, and fabric and employed Ansible YAML playbooks and Docker Python scripts as robust

- Developed comprehensive functional and load tests for web interfaces using Playwright and Locust Python scripts.
- Diagnosed inefficiencies and gaps in existing ecosystems and recommended/implemented remediation through process improvement, automation and tooling.
- Engaged in agile ceremonies including planning, retrospectives, backlog refinement, demonstrations and daily stand-ups.
- Conducted code reviews, provided technical guidance, and mentored junior developers to improve code quality and development practices.

Career Break (Jan. 2019 - Nov. 2019)

This period was taken to renovate the house and relocate

Qualcomm, Automation Engineer (Aug. 2016 – Dec. 2018)

- Resolved issues within the Python framework and test code; identified and remedied bugs discovered. Engineered infrastructure automation software following CI/CD principles.
- Developed and implemented a strategy for converting Python 2 scripts to Python 3.
- Innovated and crafted new features for the automation framework.
- Assisted colleagues in Python code development; delivered training for new users.
- Defined and engineered novel automation tools to optimise testing and debugging efficiency.
- Designed and developed various PyQt6 tools and utilities.
- Attained 100% unit test coverage leveraging pytest.
- Bolstered and expanded the Automation for Software Configuration Management, Release, and Code Control tools in a complex Engineering setting.
- Managed source code within a distributed Git/Gerrit infrastructure.
- Orchestrated the integration of diverse tools in the toolchain, elevating team velocity while upholding top-notch quality and consistency.

Cisco, SDET (DevOps) (Jul. 2011 – Aug. 2016)

- Designed, developed and maintained robust automated test suites using Python scripts, Selenium (Python bindings), and other relevant tools to validate software functionality across various platforms and environments. Created and maintained the test infrastructure, frameworks, and tools required for automated testing. This involved writing code to automate tests, ensuring they are scalable, maintainable, and integrated into the development process.
- Worked closely with developers to understand the software architecture, design, and implementation. Participated in code reviews, offered insights into testability concerns, and advocated for best practices in testing.
- Collaborated with cross-functional teams to integrate automated tests into the CI/CD pipelines using Jenkins, ensuring continuous and reliable software delivery.
- Implemented and managed configuration management tools such as Ansible playbooks and Puppet manifests to automate deployment and maintain consistency across environments.
- Developed and maintained Linux shell scripts for various automation tasks and system configurations.
- Utilised CouchDB, Memcached, Squid, ETL, ActiveMQ, Redis, and other relevant technologies to create comprehensive test scenarios and perform system integration testing of microservice architectures.
- Managed and orchestrated virtualised environments using VMware, VirtualBox, and KVM for testing purposes.
- Utilised containerization in the form of Docker to provide consistent development environments.
- Conducted performance testing using tools like JMeter, ensuring software scalability and efficiency.
- Utilised Elasticsearch Stack (ELK), Grafana, Graphite, and Prometheus for log management, monitoring, and analysis of system metrics.
- Diagnosed and remediated complex network issues with the aid of Wireshark, tcpdump, and nmap.
- Actively contributed to the identification and implementation of best practices for software testing and quality assurance. Project management and bug tracking were accomplished with the aid of Jira.

Solarflare Communications, SDET (Mar. 2010 – Jun. 2011)

- Engineered and implemented an automated test framework in Python, integrating open source and commercial tools to validate the performance and protocol functionality of 10Gb network devices.
- Conducted meticulous comparative performance tests against competitors' products, leveraging Matplotlib to visualise and interpret the results graphically.
- Crafted test specifications based on IETF RFC and IEEE standards, facilitating protocol verification by automating processes with Ixia packet generators.
- Orchestrated the creation and maintenance of reporting dashboards using Django.
- Resolved intricate automation and product-testing challenges, collaborating closely with cross-functional product teams to deploy robust test strategies, plans, and processes.

- Generated product-specific test plans, ensuring comprehensive coverage aligned with precise product requirements.
- Designed and developed web scraping scripts to extract pertinent competitor product data.

3Com, QA Automation Engineer *(Feb. 2009 – Jan. 2010)*

- Tested all functional and non-functional aspects of C++ embedded software on a unified security platform, encompassing firewall, Router, web filtering, spam filtering and anti-Virus functionality.
- Utilised Spirent Smartbits packet generators, Avalanche traffic generators and automated tests with Python, Perl and Tcl.

Citrix, QA Team Leader *(Dec. 2005 – Dec. 2008)*

- Completed many diverse full life cycle testing projects, including release 4.5 of Presentation Server, XenApp, XenDesktop, XenServer and Edgesight products.
- Complex server farm environments were generated using VMware with automation scripts.
- Collaborated with product teams to implement and deploy test strategies, test plans, and test processes.

Additional Experience

- Network Support Analyst at BT, responsible for 3rd line support of ISP Network infrastructure.
- Network protocol verification engineer at 3Com, designing test specifications and maintaining test automation systems for the verification of networking protocols with switch and router products.

Security Clearance

- SC (Security Check): Current.
- BPSS (Baseline Personnel Security Standard): Current
- DBS (Disclosure and Barring check): Current

Education

- MSc. Biochemistry, University of Reading.
- BSc. Biochemistry with Chemistry, University of Reading.
- 'A' Levels: Physics, Chemistry, Pure Mathematics and Biology.

Professional Certifications

- Cisco: CCNA, CCNP, CCDA, CCDP
- Citrix: CCA, CCEA

Projects

Commercially viable IoT product

- Designed from scratch the hardware requirements and software for an original IoT project.
- Investigated the many aspects of commercial IoT projects, including legal/patent and manufacturing requirements. Conducted performance testing of numerous single-board computers to evaluate computational and power requirements.
- Fabricated a final fully operational prototype demonstrator utilising OpenCV for image recognition along with a Django management interface. Styling was implemented using Tailwind CSS.
- Implemented RabbitMQ for front-end and back-end separation.
- Telemetry was provided by MQTT with a Mosquitto broker, together with Twilio SMS alerting.
- Monitored and stored sensor data in InfluxDB, visualised through Grafana.

Full career history available upon request.

Latest CV: https://raw.githubusercontent.com/stephen-ra-king/cv/main/Stephen_King_CV.docx

Version: 0.1.21